
FORMULA & MODEL DOCUMENTATION

TN SENTINEL X PRO

Data Preparation

- Imported raw dataset
 - Cleaned using Microsoft Power Query
 - Removed duplicates
 - Standardized values
 - Created analytical model
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Calculated Metrics

The following calculated fields were developed:

- Risk Score
 - Risk Category
 - District Rank
 - Women Safety Ratio
 - Murder Rate
 - Accident Rate
 - Severity Index
 - Public Safety Index
 - Crime Contribution Percentage
 - Priority Level
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Excel Functions Used

- XLOOKUP
- IF

- IFS
 - INDEX
 - MATCH
 - ROUND
 - AVERAGE
 - MAX
 - MIN
 - SUM
 - COUNTIF
 - RANK
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Excel Features Used

- Power Query
 - Pivot Tables
 - Pivot Charts
 - Slicers
 - Conditional Formatting
 - Data Validation
 - Named Tables
 - Named Ranges
 - Hyperlinks
 - Workbook Protection
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VBA Automation

The workbook includes VBA automation for:

- Dashboard navigation
- Dynamic KPI card color updates
- Interactive workbook navigation

Dashboard Components

Executive Dashboard

- KPI Cards
- Risk Analysis
- Crime Distribution
- District Rankings
- Women Safety Analysis

District Intelligence

- District Dossier
- Risk Assessment
- Crime Composition
- District Comparison
- Recommendation Engine

Executive Insights Report

- Executive Summary
- Key Findings
- Strategic Insights
- Strategic Recommendations
- Project Conclusion